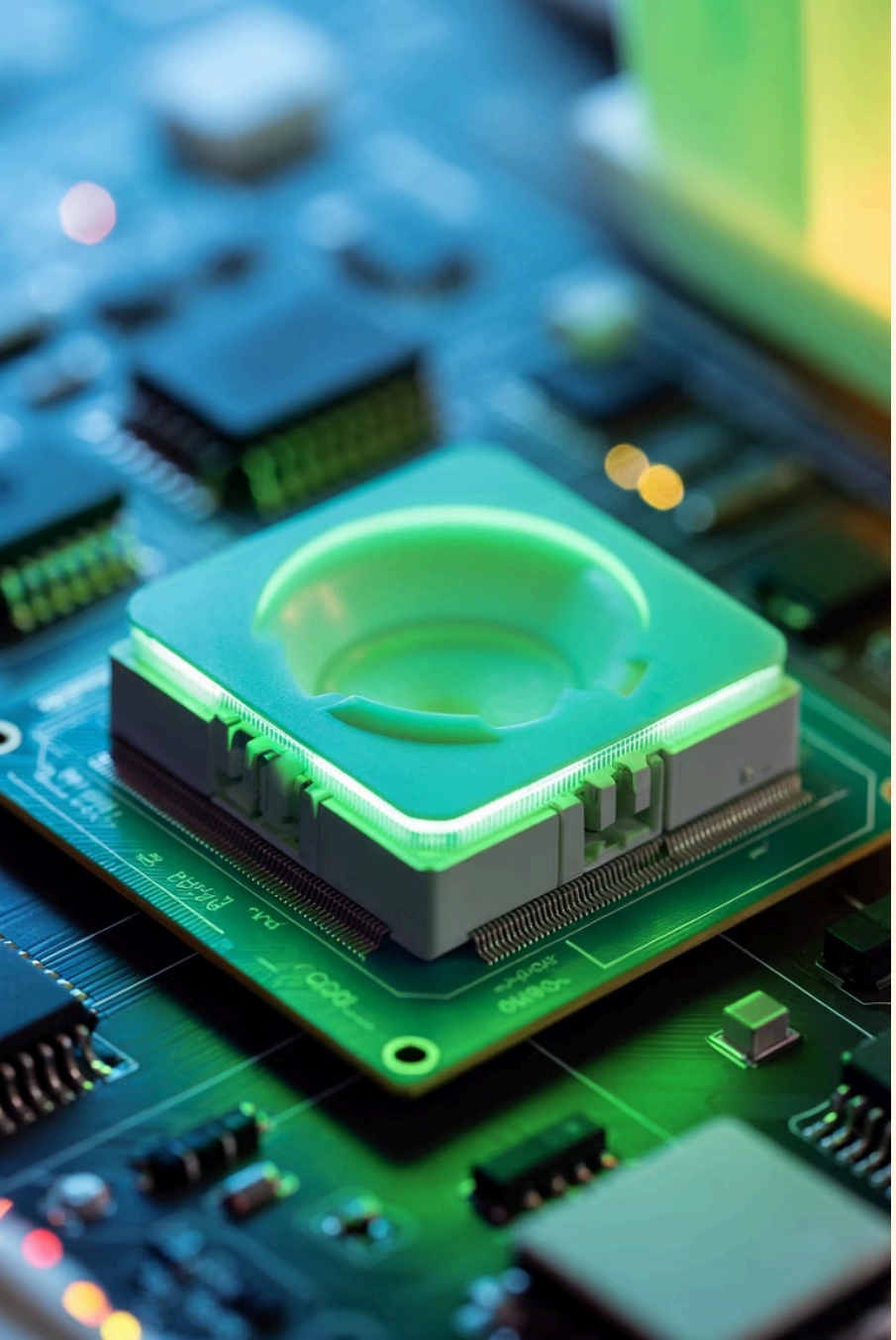


AI-Based Smart Ultrasonic Radar System

Design Thinking Engineering Project

Aadhavan K

Made with GAMMA

Empathize: Real-World Scenarios

Blind Spot Detection

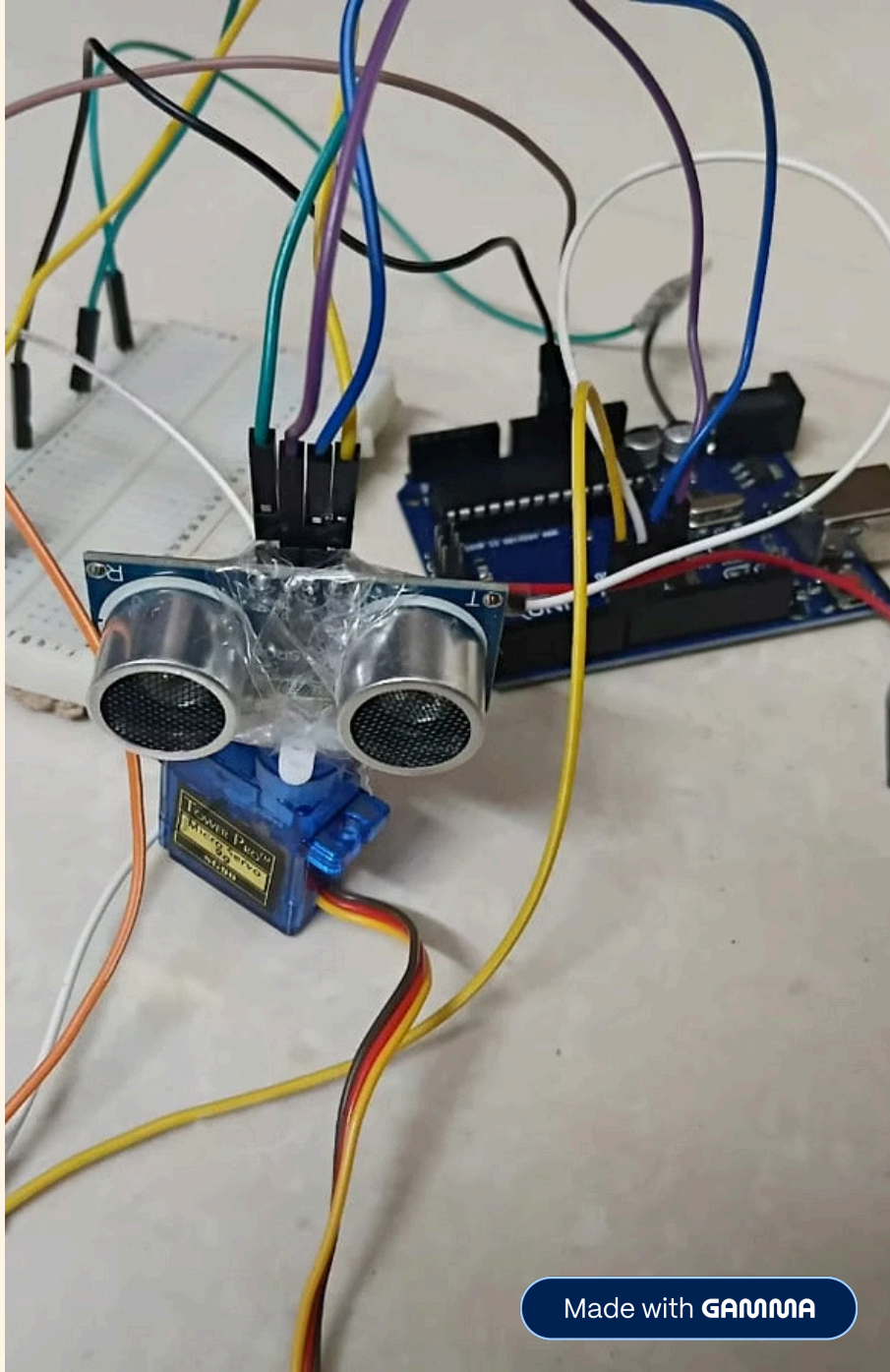
Vehicles and robots struggle with unseen obstacles in critical zones

Low Visibility

Dark environments, fog, and adverse weather conditions limit detection

Obstacle Detection

Real-time identification needed for safety and navigation systems



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Empathy Map

Think

"Is there something behind me?"

"Can I trust this system?"

Feel

Anxious about unseen dangers

Relieved with reliable alerts

Need

Real-time detection

Accurate distance measurement

Do

Check mirrors repeatedly

Depend on sensor feedback

User: Driver, Robot Operator, Security System

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Define: Problem Statement

> How might we detect obstacles reliably in darkness, fog, or low-light conditions without expensive equipment?

Point of View

Drivers and autonomous systems need affordable, reliable obstacle detection that works in any lighting condition

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Ideate: Possible Solutions

1

Camera Vision

Image processing with computer vision algorithms

2

LIDAR System

High-precision laser distance measurement

3

Infrared Sensors

Heat-based detection for night vision

4

Ultrasonic Radar

Sound wave-based distance measurement

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Final Idea Selection

Why Ultrasonic Radar?

- **Low cost** - minimal hardware investment
- **Works in dark** - independent of light
- **Weather resistant** - reliable in fog
- **Efficient** - fast response time

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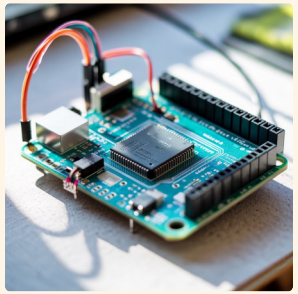
Prototype: Hardware Components



Arduino Uno

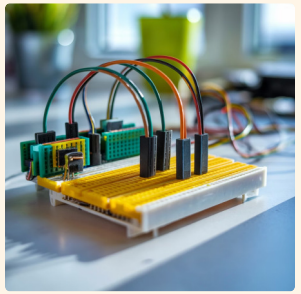
Microcontroller for processing

HC-SR04 Sensor

Distance measurement

Servo Motor

360° scanning capability



Breadboard

Prototyping connections

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Prototype: Working Flow

Signal Transmission

Ultrasonic sensor emits sound waves

Object Reflection

Waves bounce off obstacles

Time Calculation

Distance = (speed × time) / 2

Servo Rotation

Motor scans 360° field

Data Display

Radar visualization on interface

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Prototype: AI Logic & Output

Distance Classification

Safe (>100cm)

Green indicator

Warning (50-100cm)

Yellow alert

Critical (<50cm)

Red alarm

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Testing & Results

Distance Testing

Measured accuracy across 10-200cm range

Moving Objects

Verified real-time tracking capability

Angle Coverage

360° scanning with 1° precision steps

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